

Name of project :

YAFURAH II – Control System - HW

Factory Acceptance Test (FAT)

The FAT shall be performed by the vendor and will include the following activities(Test schedule):

Item	Description	Remark
1	Initial meeting (FAT document and plan review, scope of tests, schedule, etc.)	
2	Vendor documentation (including in-house test reports) check	
3	Hardware and software inventory check	
4	Mechanical inspection	
5	Wiring and termination inspection	
6	Start-up test and general system functions	
7	System alarm test	
8	System alarm test	
9	Operator interface	not applicable in this part
10	IO tests from field wiring terminals to I/O terminals, to controller and including HMI.	not applicable in this part
11	Complex functionality and operation modes (for example, batch, sequence control)	not applicable in this part
12	Test of communication links to subsystems and third-party interfaces	
13	FAT rework, punch list development	
14	FAT close-out meeting	

AUTEL a.s.

05-08-2026

OK

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FAT test report

Item 2 - Vendor documentation check

Purpose

To review all FAT relevant documents.

Item	Verify document	Test result	Remarks
1		☒ P ☐ F ☐ NA	
2	Manuals	☒ P ☐ F ☐ NA	
3	Quality documentation	☒ P ☐ F ☐ NA	

Comments

Nonconformity to be recorded in the punch list, categorized and treated as agreed in the start-up meeting.

P = Pass

F = Fail

NA = Not applicable

05-08-2026

Signature:

5.8.2026

DSPW

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Item 3 - Hardware and Software inventory check

Purpose

To verify that HW architecture, quantities, dimensions, painting, etc. are in accordance with the relevant documents. Furthermore, SW licences, spares and consumables shall be checked.

Reference documents:

- Approved vendor cabinet layout drawings
- Order

Note

Recommendation: copies of the relevant drawings shall be checked off and signed by the customer and user to serve as proof.

Item	Description	Test result
1	HW check	☒ P ☐ F ☐ NA
2	Check of SW licenses /versions including firmware	☐ P ☐ F ☒ NA
3	Spares, consumables, and tools	☒ P ☐ F ☐ NA

*

Comments

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P = Pass

F = Fail

NA = Not applicable

05-08-2026

Signature:

5.8.2026

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*The purpose of testing the I/O modules is only to verify that the connection from the terminals to the I/O cards of the control system is correct.

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Item 4 - Mechanical inspection

Purpose

To inspect HW architecture and layout as against the approved documents.

Reference documents:

- Approved vendor cabinet layout drawings
- Requirement specification
- Applicable standards and codes

Note

Items under 4 will be randomly selected for the test.

Item	Description	Test result
1	Cable entry, support bars and accessories (cable clamps, glands, etc.)	☒ P ☐ F ☐ NA
2	Labeling, tagging	☒ P ☐ F ☐ NA
3	Mounting of components and modules	☒ P ☐ F ☐ NA
4	Screwed connections, terminal connections	☒ P ☐ F ☐ NA
5	Earthing, equipotential bonding	☒ P ☐ F ☐ NA
6	Electric shock protection, warning labels	☒ P ☐ F ☐ NA
7	Maintainability of cabinet fans, construction of cabinets	☒ P ☐ F ☐ NA
8	Spare capacity	☒ P ☐ F ☐ NA

Comments

Nonconformity to be recorded in the punch list, categorized and treated as agreed in the start-up meeting.

P = Pass

F = Fail

NA = Not applicable

05-08-2026

OK

Signature:

5.8.2026

DSPW

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[Signature]

Item 5 - Wiring and termination inspection

Purpose

To verify that the wiring is in accordance with the guidelines given by the requirement specification and that the approved HW documents and workmanship are in compliance with industry standards.

Reference documents

- Approved vendor cabinet layout drawings
- Requirement specification
- Applicable standards and codes

Note

Items 5 and 6 will be randomly selected for test.

Item	Description	Test result
1	Wiring and cabling, cabling of internal circuits	☒ P ☐ F ☐ NA
2	Fusing, circuit-breakers	☒ P ☐ F ☐ NA
3	Tagging, labelling	☒ P ☐ F ☐ NA
4	Segregation of lines, colors, cross-section, voltages , Ex-i	☒ P ☐ F ☐ NA
5	Wire crimp inspection	☒ P ☐ F ☐ NA
6	Manual wire crimp pull test	☒ P ☐ F ☐ NA
7	Cable duct loading	☒ P ☐ F ☐ NA
8	I/O wiring to terminals and connector labeling	☒ P ☐ F ☐ NA
9	System cable plug orientation	☒ P ☐ F ☐ NA
10	System voltage insulation test	☐ P ☐ F ☒ NA

Comments

Nonconformity to be recorded in the punch list, categorized and treated as agreed in the start-up meeting.

P = Pass

F = Fail

NA = Not applicable

Signature:

5.8.2026

DSPW

[Signature]

[Signature]

05-08-2026

OK

[Signature]

Item 6 - Start-up test and general system functions

Purpose

To verify that the system has no problems to start up, recover from a power failure and perform on-line loading. Furthermore, it shall be verified that the system is operating within the given limits.

Reference documents

- Relevant vendor documentation
- Requirement specification (specified system limits)

Item	Description	Test result
1	New start (start from zero point ^a , stop/start)	☒ P ☐ F ☐ NA
2	On-line changes	☒ P ☐ F ☐ NA
3	Controller cycle time	☒ P ☐ F ☐ NA
4	Display call-up time	☒ P ☐ F ☐ NA
5	Value update time	☒ P ☐ F ☐ NA
6	System load (memory capacity, storage capacity, etc.)	☒ P ☐ F ☐ NA
7	Log-on strategy and level	☒ P ☐ F ☐ NA
8	Alarm processing strategy and acknowledgement	☒ P ☐ F ☐ NA
^a New disc and controller back-up batteries removed.		

Comments

Nonconformity to be recorded in the punch list, categorized and treated as agreed in the start-up meeting.

P = Pass

F = Fail

NA = Not applicable

Signature:

5.8.2026

DSPW





05-08-2026

OK 

Item 7 - System alarm test

Purpose

To verify the announcement of system-related failures, cabinet alarms and system-generated alarms to appear on system.

Reference documents

- Vendor documentation
- HMI specification

Item	Description	Test result
1	Power-supply failure, UPS monitoring	☒ P ☐ F ☐ NA
2	Fuse, breaker monitoring	☒ P ☐ F ☐ NA
3	Cooling fans	☒ P ☐ F ☐ NA
4	Communication, network monitoring	☒ P ☐ F ☐ NA
5	Short circuit, wire break, out of range, earth fault	☒ P ☐ F ☐ NA
6	Watchdog, if any	☒ P ☐ F ☐ NA

Comments

Nonconformity to be recorded in the punch list, categorized and treated as agreed in the start-up meeting.

P = Pass

F = Fail

NA = Not applicable

05-08-2026

Signature:

5.8.2026

DSPW

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OK *[Handwritten signature]*

Item 8 - Hardware redundancy and diagnostic

Purpose

To ensure proper operation and monitoring of redundant components.

Reference documents

- Vendor relevant documentation

Note

No single action shall effect the operation of the overall system.

Item	Description	Test result
1	Redundant operation and monitoring of controllers	☒ P ☐ F ☐ NA
2	Redundant operation and monitoring of communication and networks	☒ P ☐ F ☐ NA
3	Redundant operation and monitoring of power supplies	☒ P ☐ F ☐ NA
4	Redundant operation and monitoring of operator stations	☒ P ☐ F ☐ NA
5	Redundant operation and monitoring of I/Os, if any	☒ P ☐ F ☐ NA
6	Redundant operation and monitoring of all other devices not mentioned before	☐ P ☐ F ☒ NA

Comments

Nonconformity to be recorded in the punch list, categorized and treated as agreed in the start-up meeting.

P = Pass

F = Fail

NA = Not applicable

05-08-2026

Signature:

5.8.2026

DISPW

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OK *[Handwritten signature]*

Item 12 - Test of communication links to subsystems and third-party interfaces

Purpose

To verify the interoperability of the systems considered.

Reference documents

- I/O list for relevant signals
- Relevant function diagram or logic diagram, C&E, etc.

Note

The majority are serial links (soft I/O). Test shall be performed as with hardwired I/O after consideration of the side condition of serial links, for example, transmission time.

Item	Description	Test result
1	Detailed check of functionality	☐ P ☐ F ☒ NA

Comments

Nonconformity to be recorded in the punch list, categorized and treated as agreed in the start-up meeting.

P = Pass

F = Fail

NA = Not applicable

Signature:

5.8.2026

DSPW

CD

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05-08-2026

OK - Ray -

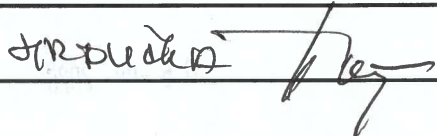
FAT certificate

☒ ACCEPTED

☐ NOT ACCEPTED

CUSTOMER	DOOSAN ŠKODA POWER s.r.o. Tylova 1/57, 301 28 Plzeň, Česká republika		
PROJECT	YAFURAH II	PROJECT NO	P100325
PLANT/UNIT	Harmaliyah GOSP 1, Kingdom of Saudi Arabia		
VENUE OF FAT	Autel,a.s. Oldřichovice 790, 739 01, Třinec, Česká republika	FAT FINISHED ON	5.8.2026

PERSON IN CHARGE/ Signatures:

CUSTOMER		DEP.	DSPW
CUSTOMER		DEP.	OB Dusan
CUSTOMER		DEP.	
VENDOR	PETRA KOJNAK 	DEP.	AUTEL a.s.
VENDOR	PETR FUKALA 	DEP.	AUTEL a.s.
VENDOR	Roman Filla 	DEP.	Autel, a.s.

[illegible]

FAT PUNCH LIST

Any incomplete work or nonconformities shall be recorded on the FAT punch list and categorized as follows:

- **Immediate** – FAT to be paused for rectification, FAT to continue after rectification;
- **On-going** – Rectification during FAT, will be completely rectified by end of FAT (typically for multiple instances of the same issue, each of which will be rectified as encountered and by the end of the FAT, all will be rectified. Rectification is performed before test. If later re-test is required, the item is classified as "Repeat");
- **Repeat** – FAT to be repeated for this item after rectification and before completion of FAT;
- **Before shipment** – To be rectified after FAT but before the system is shipped to the site, re-test, if any, to be documented and conducted during SAT;
- **At site** – Remaining work to be rectified after shipment, i.e., at site, and re-test, if any, to be documented and conducted during SAT or SIT; or
- **Future** – To capture issues not in the project or FAT scope but something to be addressed in the future.

Note

NO.	Reference to the applicable specification	Description of the issue	Description of the resolution and date completed	Acceptance (signature of buyer & vendor)
1	As per OBE Doosan punch list			
2				
3				
4				
5				
6				
7				
8				
9				
10				

PUNCH LIST

Project Name	Vendor / Place of Inspection	Purchase Order No.	Report No.
JAFURAH II COCGEN PROJECT	Autel a.s. / Oldřichovice 790, 739 61, Třinec, Czech republic	P100325	IR-DE-TPI-26-1449-Jafurah2
Vendor Person Contacted / Signature	Inspector Name / Signature	Item Name	Issued Date
Jan Fišer	Miroslav Hrdlička	Steam Turbine	05.08.2026

No.	Issued Date	Item No. (Doc. No.)	Punch Description	Action By	Closed Date	Remark
1	05.08.2026	D251793302.NS1 / +20CJJ01GH001R	- The cabinet layout drawing did not fully correspond to the manufactured TCS/TPS cabinet. Differences were identified in the cable duct arrangement and component layout.	Autel a.s.	05.08.2026	Drawing updated during FAT.
2	05.08.2026	D251793302.NS1 / +20CJJ01GH002F	- Component F93 was missing from the cabinet layout drawing and did not reflect the manufactured cabinet configuration.	Autel a.s.	05.08.2026	Drawing updated during FAT.
3	05.08.2026	D251793302.NS1 / +20CJJ01GH005R	- Component location in the cabinet layout drawing did not match the manufactured cabinet configuration.	Autel a.s.	05.08.2026	Drawing updated during FAT.
4	05.08.2026	+20CJJ01GH001F	- The cabinet is equipped with a French (Type E) service socket. To ensure compatibility with locally available plugs, a power adapter will be supplied.	Autel a.s.	Before shipment	Adapter will be provided.

05-08-2026

OBO Doornik
flyDSPW
yoAUTE L. a.s.
B. F.POWER ADAPTER
WAS ADDED TO
CABINET
10.08.2026
Třinec - Oldřichovice 790
739 61 Třinec 1